



Use of a spatiotemporal approach for understanding preschoolers' playground activity

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ABSTRACT

This study used spatiotemporal hot-spot analysis to characterize physical activity on the childcare center playground. Preschool-aged children ($N = 34$) wore a GPS and accelerometer during 2–3 outdoor periods on one day. A spatiotemporal weights matrix was generated so that points within a specified distance in meters (space) and 3 min (time) were considered neighbors. The Getis-Ord G^* statistic was calculated to detect locations of significant hot/cold spots in vector magnitude counts/15–sec. Hot/cold spots changed within a single outdoor period and between outdoor periods, highlighting the importance of time. This approach can be used to identify points of intervention during provided outdoor time.

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1. Introduction

It is recommended that preschool-aged children obtain at least 180 min per day of total (light-to-vigorous) physical activity (Commonwealth of Australia, 2010; UK Chief Medical Officers, 2011; Tremblay et al., 2017; United States Department of Health and Human Services [USDHHS], 2018). As 59% of children under 5 years of age attend regular center-based childcare, this is an important setting in which to accumulate physical activity (Corcoran and Katrina, 2019). While it is widely recognized that outdoor time during childcare provides a daily opportunity for free-play physical activity and promotes higher levels of physical activity compared to indoor time, preschoolers still spend 28 to 66% of provided outdoor time in sedentary behavior (Brown et al., 2009; Gubbels et al., 2011; Sugiyama et al., 2012; Trost et al., 2008). Activity levels during outdoor time vary substantially among children attending different preschools or childcare centers (Gubbels et al., 2011; Pate et al., 2004; Tandon et al., 2013; Vanderloo et al., 2014), which suggests there may be contextual factors or aspects of the environment that affect this behavior.

The physical environment has a salient influence on physical activity during the preschool day (Bower et al., 2008; Dowda et al., 2009; Sugiyama et al., 2012). In the outdoor environment, preschoolers have been shown to be more physically active on playgrounds rated as higher quality (Bower et al., 2008; Gubbels et al., 2011), that are adequately sized (Boldemann et al.,

2006; Clark et al., 2002; Dowda et al., 2009), have portable equipment (Berg, 2015; Dowda et al., 2009; Gunter et al., 2012), grassy spaces (Berg, 2015), or fixed equipment (Gunter et al., 2012; Sugiyama et al., 2012). Many of these studies have involved correlating a variable (e.g., total area, availability of a slide, number of pieces of portable equipment) with overall physical activity levels. Fewer researchers have investigated where preschoolers actually spend their outdoor time and how physically active they are in various locations within the playground. This approach can inform us as to what features of the playground children are actually using during outdoor time overall and while being physically active.

The available literature in preschool-aged samples supports that physical activity varies with a child's location on the playground (Cosco et al., 2010; Hustyi et al., 2012; Larson et al., 2014; Nicaise et al., 2011; Smith et al., 2014) and that modification of these locations (e.g., addition of a slope to a grassy field) can promote increases in physical activity participation (Nicaise et al., 2012). Further, Smith et al. (2014) reported that 23% of the variance in children's physical activity could be explained by attributes of the child's location within the playground (e.g., open area, sand area). Studies of older children and adolescents note that in some locations within the playground, the commonly seen disparity in physical activity by sex is reduced (Andersen et al., 2019; Black et al., 2015; Clevenger et al., 2018; Howe et al., 2018; Saint-Maurice et al., 2011; Willenberg et al., 2010). Thus, information about playground locations that promote physical activity is

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needed and could inform the design of playgrounds and interventions that promote physical activity in all children, including those who typically demonstrate lower physical activity levels (e.g., girls).

While research on the locational context of preschoolers' physical activity patterns is warranted, measuring physical activity and its context is difficult due to the complexity and brevity of children's movements (Oliver et al., 2007). Direct observation can provide context to physical activity, but it is difficult to obtain continuous, individual-level data using this time-consuming method that depends heavily on researcher reliability (Oliver et al., 2007). Monitor-based approaches can provide continuous, individual-level data on the intensity, duration, and location of children throughout the day. A few studies have used Global Positioning Systems (GPS) receiver units in combination with accelerometers (Andersen et al., 2019; Andersen et al., 2015; Clevenger et al., 2018; Pawlowski et al., 2016) or heart rate monitors (Fjørtoft et al., 2009, 2010). However, this method has not been used in preschools, where playgrounds may be smaller and denser. A variety of approaches for analyzing these data have been used, including visualization (e.g., grid maps; Fjørtoft et al., 2010) and statistical approaches (e.g., hot spot analysis; Andersen et al., 2019; Pawlowski et al., 2016) (Clevenger et al., 2018). While the development of a consistent method is critical to ensure comparability between studies, the methods used to date have one major limitation. That is, they do not allow for spatially relevant information to vary over time. For example, hot spot analysis has been used to identify clusters of high accelerometer counts (Andersen et al., 2019; Clevenger et al., 2018; Pawlowski et al., 2016), but it does not identify whether these clusters are present over the entire outdoor period, or just during a particular time (e.g., at the beginning of outdoor time). McKenzie et al. (1997) and Holmes (2012) demonstrated that physical activity participation varies over the course of the outdoor period, so it is likely that children's location would also vary over time. A spatiotemporal method could be used to identify what areas of the playground elicit high levels of physical activity at various time points during the outdoor period.

The purpose of this study was to use a spatiotemporal approach to hot spot analysis to identify where and when clusters of physical activity occur on the playground. We hypothesized that clustering would be present, but the location of clusters would change within the course of a given outdoor period (intraperiod), or between outdoor periods (interperiod).

2. Materials and methods

2.1. Participants and setting

Licensed childcare centers were recruited and informed consent forms were sent home to parents of children in classrooms containing 2–5 y olds. Prior to participation, each child provided verbal assent. Children with no apparent disease or condition that would alter physical activity participation were eligible to participate. On the day of data collection, weather (e.g., temperature) and provided equipment (e.g., the teacher brought out hula hoops) were recorded. For each outdoor period, prior deprivation was recorded (amount of time since the last outdoor play period), as previous research has indicated this may impact physical activity participation (Pellegrini et al., 1995; Smith and Hagan, 1980). The University's Institutional Review Board approved this protocol.

2.2. Measurement of physical activity

Children wore an Actigraph wGT3X-BT accelerometer (ActiGraph LLC, Pensacola, FL) at the right hip, secured by an elastic belt, for one entire day at the childcare center. The Actigraph

wGT3X-BT is a triaxial accelerometer that measures raw acceleration (g's) ranging from ± 8 g's in three planes (vertical, antero-posterior, and mediolateral), which is filtered and summed over a user-specified epoch to create activity counts. Accelerometers were initialized to collect raw acceleration at a sampling rate of 30 Hz and data were downloaded as re-integrated counts using ActiLife software (version 6.13.3). Vector magnitude (the square root of the sum of the squared activity counts from each axis) was calculated for each 15-s epoch. Vector magnitude was used because it is an orientation-independent metric, and a 15-s epoch was used as it is the most commonly used epoch length in studies of preschool-aged samples (Miguel et al., 2017).

2.3. Measurement of playground location

Each child also wore a QStarz BTQ1300ST GPS receiver unit (QStarz International Co., Ltd., Taipei, Taiwan) secured to the same elastic belt as the accelerometer. Location (as latitude and longitude) was recorded at a 5-s epoch and relevant almanac and ephemeris data were downloaded to each GPS device prior to data collection to decrease time to initial signal acquisition. This monitor has been used previously in the playground setting (Andersen et al., 2015; Pawlowski et al., 2016) and in a review of 6 portable GPS receivers, the QStarz was more sensitive and accurate, operated on more channels, acquired a signal faster (from a cold, warm, and hot start), and stored more data points than the other monitors (Duncan et al., 2013). The manufacturer-reported error for this unit is <3 m (QStarz

). Data were downloaded using the QSports software (version 3.76) and exported as a 'csv' file.

2.4. Data analysis

To detect potential environmental factors that would cause additional GPS error, we used Trimble Global Navigation Satellite System (GNSS) Planning Online software (version 1.2.0.0, Trimble Terrasat GmbH, Germany) to identify positional dilution of precision (PDOP), ionospheric index (including potential scintillation and total electron content), and number of satellites in view on the dates and times of measurement. The Ap index, as an indicator of planetary geomagnetic disturbance, was obtained from the German Research centre for Geosciences. All metrics were within accepted ranges (e.g., PDOP was <2 , ionospheric index of 1, ≥ 6 satellites in view, and $Ap \leq 8$), indicating higher-than-expected positional error was unlikely.

Workflow is diagrammed in Fig. 1. For both accelerometer and GPS data, outdoor time was isolated prior to analysis based on start and end times determined using video observation of the playground. Accelerometer and GPS data were matched based on time stamp in R (R Core Development Team, Vienna, Austria). Because the GPS unit does not allow the user to set a specific start time, there were some GPS time stamps that did not directly match accelerometer time stamps (e.g., 12:10:01 vs. 12:10:00). Exact matches between the two data sets were completed first; then for accelerometer data points without a corresponding GPS point, latitude and longitude were assigned based on the closest GPS point, as long as the two time stamps were within 1 min of each other. The resulting dataset consisted of vector magnitude counts and a corresponding GPS-determined location for each 15-s epoch for each child.

These data were combined for each class (group-level analysis) and were loaded into ArcMap (ESRI, Redlands CA, version 10.5.1) and converted to the WGS 1984 UTM Zone 16 N projected coordinate system. Global Moran's I was used to assess whether significant global (overall) clustering was present in the data (Moran, 1950). Similar to previous research (Clevenger et al.,

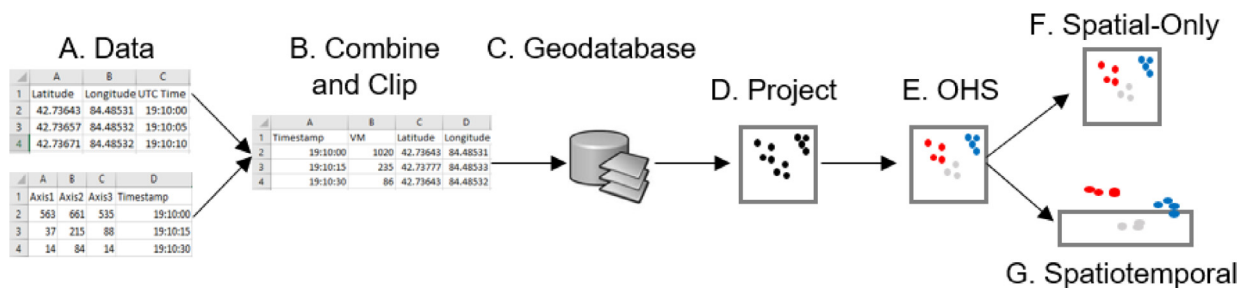


Fig. 1. Diagram of workflow for analyzing GPS plus accelerometer data. GPS data were cleaned (A) and combined with accelerometer data for each 15-s epoch based on time-stamp, and non-outdoor times were removed (B). Files were stored in a geodatabase (C) and then projected in ArcMap (D). Optimal hot spot analysis (OHS) was used to determine the distance band (E) that was subsequently used for the spatial-only hot spot analysis (F) and spatiotemporal hot spot analysis (G).

Table 1

Descriptions of each playground.

Playground	Description of settings	Description of use
1	Mulch surface enclosed by a fence with one large piece of fixed equipment at the center. Four small, spring-based pieces of equipment, a large teeter-totter, and a spinning net for climbing were also present.	Classes 1 and 2 used this classroom every day for the first and last recess periods of the day, and often, for the pre-lunchtime recess period as well
2	Larger than playground 1, and covered in grass (some slope), save for a large, open sandpit. There was also a small seating area in the northeast corner, primarily used by teachers.	Occasionally used by class 2, and more rarely (not in the present study period), class 1 (based on teacher's choice and center schedule)
3	A certified 'nature explore classroom.' Three primary areas that could be closed off by teachers when classes needed to be separated. The grassy area was primarily open, save for some portable logs and planks. The sandbox area was large, with a play kitchen and numerous toys. The fixed equipment area was comprised of rubber flooring with one large piece of fixed equipment. Additionally, a small house and teepee were present.	Used by classes 4 and 5 twice a day, and class 3 once per day. Teachers would sometime block off one portion, which separated the fixed equipment from the sandbox and grass area.
4	A large (3-acre) park used also for the oldest preschoolers, comprised of a seating area (picnic table), large grassy areas, fixed equipment (including swings) on mulch, and linear sidewalks that children primarily used for drawing with chalk. Teachers would typically bring 1–2 balls or other portable equipment.	Used by class 3 once per day, and rarely (depending on center schedule) used by class 5. Children had to walk (approximately 5 min) to use this playground.

2018), the Optimized Hot Spot Analysis tool in ArcMap (ArcMap) was used to determine the distance band within which features would be considered neighbors. This value was allowed to vary between playgrounds due to their varying sizes (i.e., on a larger playground, neighbors may be farther apart than on a smaller playground).

Hot spot analysis is used to determine statistically significant clusters of high (hot spot) or low (cold spot) values of some attribute, which, in this case, was vector magnitude counts/15-sec. For each classroom on each playground, two final hot spot analyses were conducted- 1) spatial-only and 2) spatiotemporal. For the spatial-only models, the distance band determined by the Optimized Hot Spot Analysis tool (7–10 m depending on the playground) was used with inverse distance weighting, so that features within the distance band were considered neighbors, but features closer together were conceptualized to be more similar than features further apart. For the spatiotemporal models, a spatiotemporal weight matrix was generated, using the determined distance band (7–10 m) and allowing features within three minutes of each other to be considered neighbors. While further research may be done to compare the impact of using different distance bands or time windows in this setting, combinations of 1–10 min and 3–10 m were explored to ensure that the clusters locations were consistently found and not simply an artifact of our chosen distance band and time window. Using this spatiotemporal model, features had to be neighbors both spatially and temporally to be included as part of a hot or cold spot. For all hot spot analyses, the Getis-Ord G_i^* statistic (Getis and Ord, 1992) was used to determine significant ($p < 0.05$) hot and cold spots, and the false discov-

ery correction was used to account for multiple comparisons and autocorrelation (Caldas de Castro and Singer, 2006). For both spatial and spatiotemporal models, mean vector magnitude within hot and cold spots was determined.

3. Results

Two licensed childcare centers volunteered to take part in this study (two classes at one center and three classes at the other center; $n = 17$ from each center). These two childcare centers utilized a total of four playgrounds, which are described in Table 1. Each childcare provided children with multiple (2–3) outdoor periods per day, allowing us to identify temporal patterns both within outdoor periods (intraperiod) and across outdoor periods (interperiod) within a single day. All measurement days occurred in the summer when mean temperatures ranged from 26 to 29 Celsius. One child was absent during the morning outdoor period at one center, but otherwise, all children were present for all outdoor periods, with an average daily outdoor time of 123 min/day (range 94–169 min). These outdoor periods were part of the regular classroom schedule and no changes to scheduling or timing of these outdoor periods were imposed by our study. Descriptors about outdoor period duration, the amount of prior deprivation (indoor time), and classroom activities that took place before the outdoor period are described in Table 2. For exploratory purposes, trends and averages in vector magnitude counts/15-s over the course of each outdoor period are also described in Table 2.

Data loss was approximately 5% (e.g., due to no signal acquisition). There was evidence of global clustering at all playgrounds

Table 2
Vector magnitude during each outdoor period.

Class, Day of Week, Date	Outdoor Period	Duration (min)	Prior Deprivation (hours)	Preceding Activities*	Vector Magnitude (counts/15-s)		
					Min-max	Average	Trend Notes [†]
Class 1 Thurs. June 28	Morning	56	–	Arrival at the center	0–3617	354	No change
	Lunch	20	2.0	Large and small group	0–4209	455	Slight increase
	Afternoon	69	4.0	Free-choice and snack	0–3930	809	Steady increase
Class 2 Wed. July 11	Morning	18	–	Large and small group	0–5315	699	No change
	Afternoon 1	60	1.5	Lunch and rest	0–7214	662	Slight increase
	Afternoon 2	91	2.0	Large and small group	0–2527	629	Slight decrease
Class 3 Wed. July 25	Morning	28	–	Snack and group	0–1784	583	Steady decrease
	Afternoon	66	3.0	Small group and quiet play	0–3518	562	No change
Class 4 Tues. July 31	Morning	52	–	Snack and small group	0–3643	594	No change
	Afternoon	59	5.0	Nap and snack	0–2662	576	Slight increase
Class 5 Wed. Aug 1	Morning	47	–	Arrival at the center	0–2070	407	Slight decrease
	Afternoon	47	3.0	Nap and snack	0–2209	580	Steady decrease

*Preceding activities refers to the activity scheduled for that class immediately prior to the outdoor period. Large and small group can refer to a variety of activities depending on the center, teacher, and day (e.g., reading a book together, learning about a science topic, doing an art project). Free-choice refers to time in the classroom in which children can choose where and with whom they want to play. Rest or quiet play is time when children are encouraged to be quiet but do not necessarily have to lay down for a nap.

[†] Trends were determined visually by plotting VM over the course of the outdoor period.

Table 3
Percent of intervals classified as hot or cold spot and mean vector magnitude for spatial only or spatiotemporal hot spot analyses.

		Spatial Only Model		Spatiotemporal Model	
		Percent of intervals	VM	Percent of intervals	VM
Class 1, yard 1	Hot	42.4	652 ± 516	25.4	807 ± 558
	Cold	26.2	115 ± 294	27.0	113 ± 276
Class 2, yard 1	Hot	21.6	866 ± 902	16.5	1117 ± 1128
	Cold	10.0	418 ± 397	13.4	368 ± 365
Class 2, yard 2	Hot	40.0	837 ± 493	19.2	966 ± 528
	Cold	26.4	360 ± 378	20.2	275 ± 331
Class 3, yard 3	Hot	0.0	–	2.6	1000 ± 352
	Cold	0.0	–	4.1	344 ± 290
Class 3, yard 4	Hot	15.7	965 ± 528	14.7	1035 ± 551
	Cold	53.2	415 ± 454	25.6	320 ± 444
Class 4, yard 3	Hot	25.0	745 ± 548	10.9	888 ± 615
	Cold	8.0	299 ± 352	8.0	299 ± 352
Class 5, yard 3	Hot	23.8	759 ± 406	16.7	784 ± 381
	Cold	23.0	251 ± 289	51.0	214 ± 300

during all measurement periods (all $p < 0.0001$). Distance bands used for hot spot analysis ranged from 7 to 10 m and generally increased with overall playground size. In the spatial-only hot spot analysis, 22 and 21% of features, on average, were classified as significant ($p < 0.05$) hot and cold spots, respectively, while 15 and 21% were classified as hot and cold spots, respectively, when using a spatiotemporal approach (Table 3). Average vector magnitude within hot spots was 689 vs. 942 counts/15-s and 310 vs. 276 counts/15-s within cold spots for spatial-only and spatiotemporal approaches, respectively (Table 3).

Figs. 2–6 illustrate the results of both spatial only (right-most panel) and spatiotemporal (two left-most panels) hot spot analyses for each outdoor period. Spatial-only analyses revealed that hot spots were often located on the fixed equipment (Figs. 2–6) or grass (Figs. 3 and 6) and cold spots located in the mulch (Figs. 2 and 3), sandbox (Figs. 3 and 4), or on the sidewalk (Fig. 4). One class had no spatial-only clustering present (Fig. 4, morning recess), but there were small spatiotemporal clusters present on the sandbox and in the mulch (Fig. 4).

Figs. 2–6 also illustrate the locations of spatiotemporal hot and cold spots over the course of each outdoor period ($p < 0.05$), with time on the y-axis (i.e., as you move vertically, more time has elapsed). Hot and cold spots could occur in the same location depending on the time period. For example, in Fig. 2, both hot and cold spots were located on the fixed equipment, with no definitive clustering at the beginning of morning recess, followed by a hot

spot, and then a cold spot at the beginning of the lunch recess, followed by a hot spot. During afternoon recess, there was only a hot spot in this location. In this same class, there was a persistent cold spot in the mulch in the morning recess only.

4. Discussion

The use of GPS plus accelerometry is a promising methodology for understanding where and how actively youth are playing during provided outdoor time. A number of methods have been used to analyze GPS plus physical activity data, and while it was recently concluded that one promising method is hot spot analysis (Clevenger et al., 2018), the lack of incorporation of temporal information when modeling spatial relationships is a serious limitation of studies that have used the hot spot method in this setting (Andersen et al., 2019; Clevenger et al., 2018; Pawlowski et al., 2016). As such, we demonstrated the use of hot spot analysis using spatiotemporal weight matrices and found support for our hypothesis that the location of clusters of high or low activity counts (i.e., hot and cold spots) changed both within (intrapersonal) and between outdoor periods (interpersonal).

In previous research using hot spot analysis (Andersen et al., 2019; Clevenger et al., 2018; Pawlowski et al., 2016), data were aggregated across entire outdoor periods and only spatial information was used to identify nearby neighboring features. Prior studies that included older children and/or adolescents on one play-

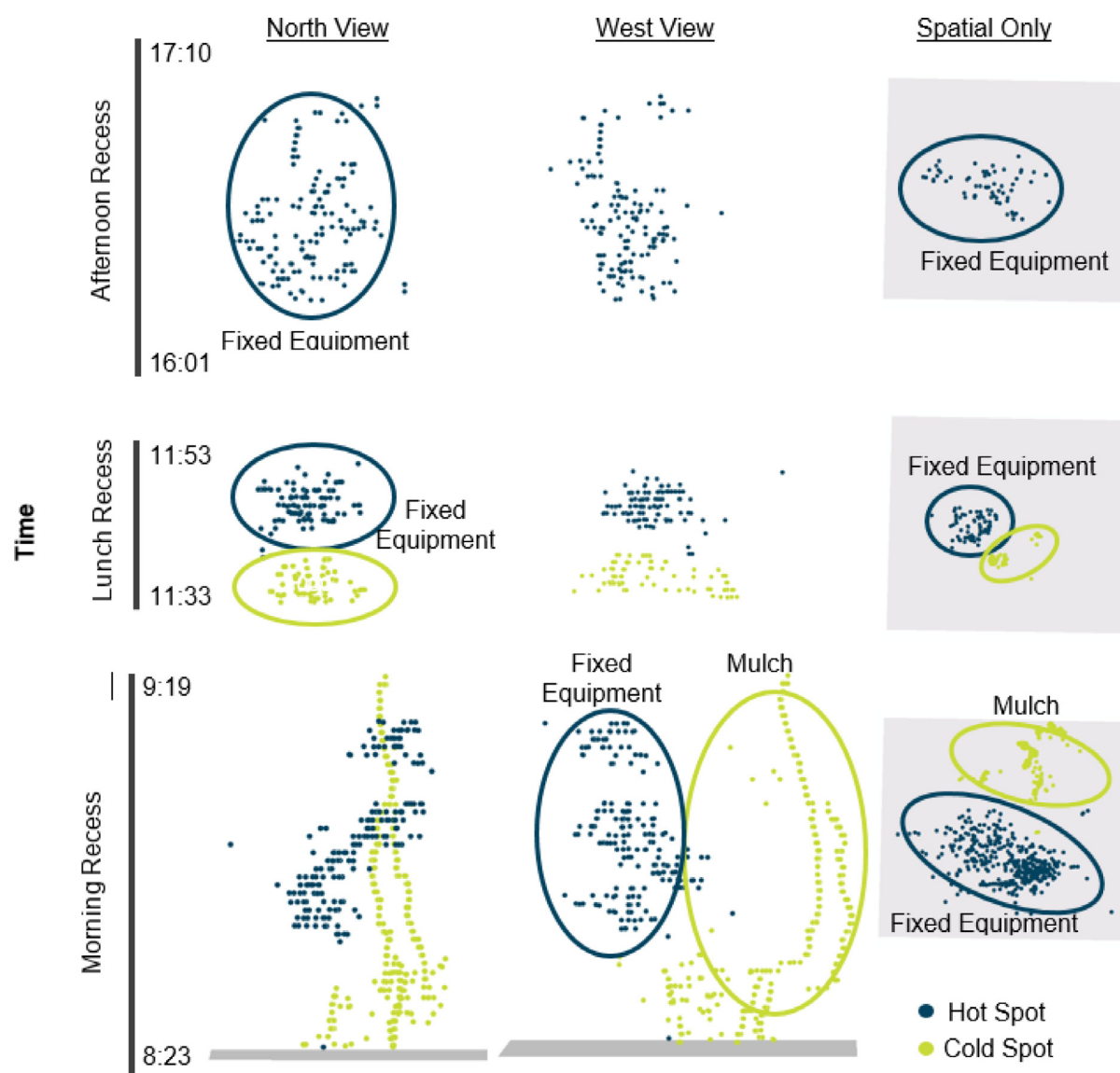


Fig. 2. Significant ($p < 0.05$) spatiotemporal hot and cold spots for classroom 1 on playground 1. For the 3-dimensional maps (two left-most panels), the “North View” shows a view from the southern portion of the playground looking North while the “West View” shows a view from the eastern portion of the playground looking West with time shown on the y-axis (further up the y-axis, more time has elapsed). The spatial only model (right-most panel) is shown as an aerial view with North at the top.

ground demonstrated hot spots of high activity counts on four square courts (Clevenger et al., 2018; Pawlowski et al., 2016), in fields (Clevenger et al., 2018; Pawlowski et al., 2016), or on fixed equipment (Clevenger et al., 2018). Similar to this previous research, we were able to identify significant clustering in preschool-aged samples on multiple playgrounds, but found that the location of hot and cold spots varied on different playgrounds (e.g., compare Fig. 2 and 3 to Fig. 4, 5, and 6). Playgrounds differ in the type, design, and size of available settings, which afford unique activities and therefore, result in varying physical activity levels, so this finding was expected. Our results support the need to use this methodology on a large number of playgrounds that vary in their design.

All but one playground had fixed equipment available, and when it was available, it typically elicited high levels of physical activity (as evidenced by the presence of hot spots), which is in line with observational and experimental research in this age group (Cosco et al., 2010; Hustyi et al., 2012; Larson et al., 2014). The only playground that had fixed equipment, but did not have a hot spot in that location was playground 3 for one class, but this

is likely related to the fact that teachers at that childcare center would often block the fixed equipment off for approximately half of provided outdoor time so that other classes could play in that area. Changes in scheduling, so that multiple classes were not outside at the same time, may promote additional physical activity in children at this childcare center, which is feasible according to a prior intervention (Van Cauwenberghe, De Bourdeauhuij, Maes, and Cardon, 2012).

Previous research has demonstrated that open spaces (Cosco et al., 2010; Nicaise et al., 2011) and sloped grass areas (Nicaise et al., 2012) promote physical activity in preschoolers, which was supported in only a third of our observations. In contrast, we found that some open spaces contained clusters of low vector magnitude counts (e.g., Figs. 2 and 3). For example, on playground 1, which was entirely comprised of fixed equipment and open mulch, children liked to pace or to stand by the fence to talk, watch cars pull in to the parking lot, or to watch children on other playgrounds play. This finding is similar to previous research in older children that demonstrated that a lack of options on the playground resulted in children congregating by a fence to

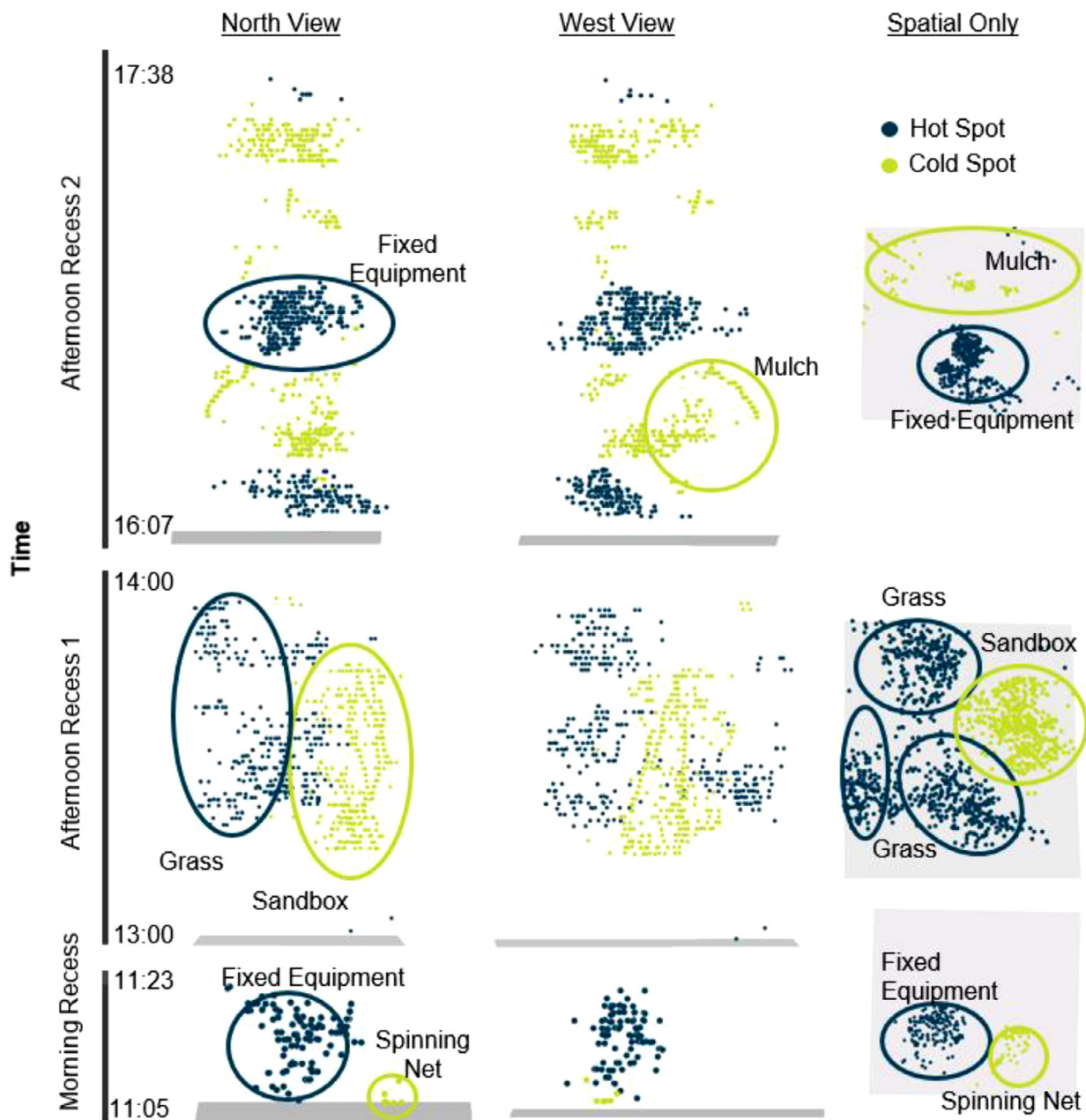


Fig. 3. Significant ($p < 0.05$) spatiotemporal hot and cold spots for classroom 2 on playground 1 (morning and afternoon 2) and 2 (afternoon 1). For the 3-dimensional maps (two left-most panels), the “North View” shows a view from the southern portion of the playground looking North while the “West View” shows a view from the eastern portion of the playground looking West with time shown on the y-axis (further up the y-axis, more time has elapsed). The spatial only model (right-most panel) is shown as an aerial view with North at the top. Figure only shows datapoints belonging to a significant hot or cold spot.

socialize (Holmes, 2012). Lack of running games or other active behaviors on the mulch may also be related to the surface type (Cosco et al., 2010), a topic that could be further explored using a combination of GPS plus accelerometry in a large number of playgrounds. Interestingly, playground 4 (Fig. 4) had a very large grassy field as it doubles as a city park, but there was not a hot spot in this area. It may be that the playground size was too large for preschoolers and therefore, they did not travel far distances during provided outdoor time. This result is also supported by the fact that preschoolers at that childcare center rarely used the swings, likely because they were sized for older children, not preschoolers. Thus, the impact of equipment and location size could be further explored. Use of GPS and accelerometry would

be able to identify, for example, changes in space use with age or motor skill development.

In addition to differences between playgrounds, we found that cluster locations could also vary on the same playground between different classes (e.g., compare Fig. 5 to Fig. 6) and within the same class during different times of the day (e.g., Fig. 2). For example, class 1 had three outdoor periods on playground 1, all with different cluster patterns. During morning recess, which occurred soon after children arrived at the childcare center, there was a prolonged cold spot in the mulch area, where children could watch other classes arrive at the center (Fig. 2). Vector magnitude was generally low during this outdoor period (Table 2). In all three time periods, there was a hot spot in the fixed equipment area, but this was in-

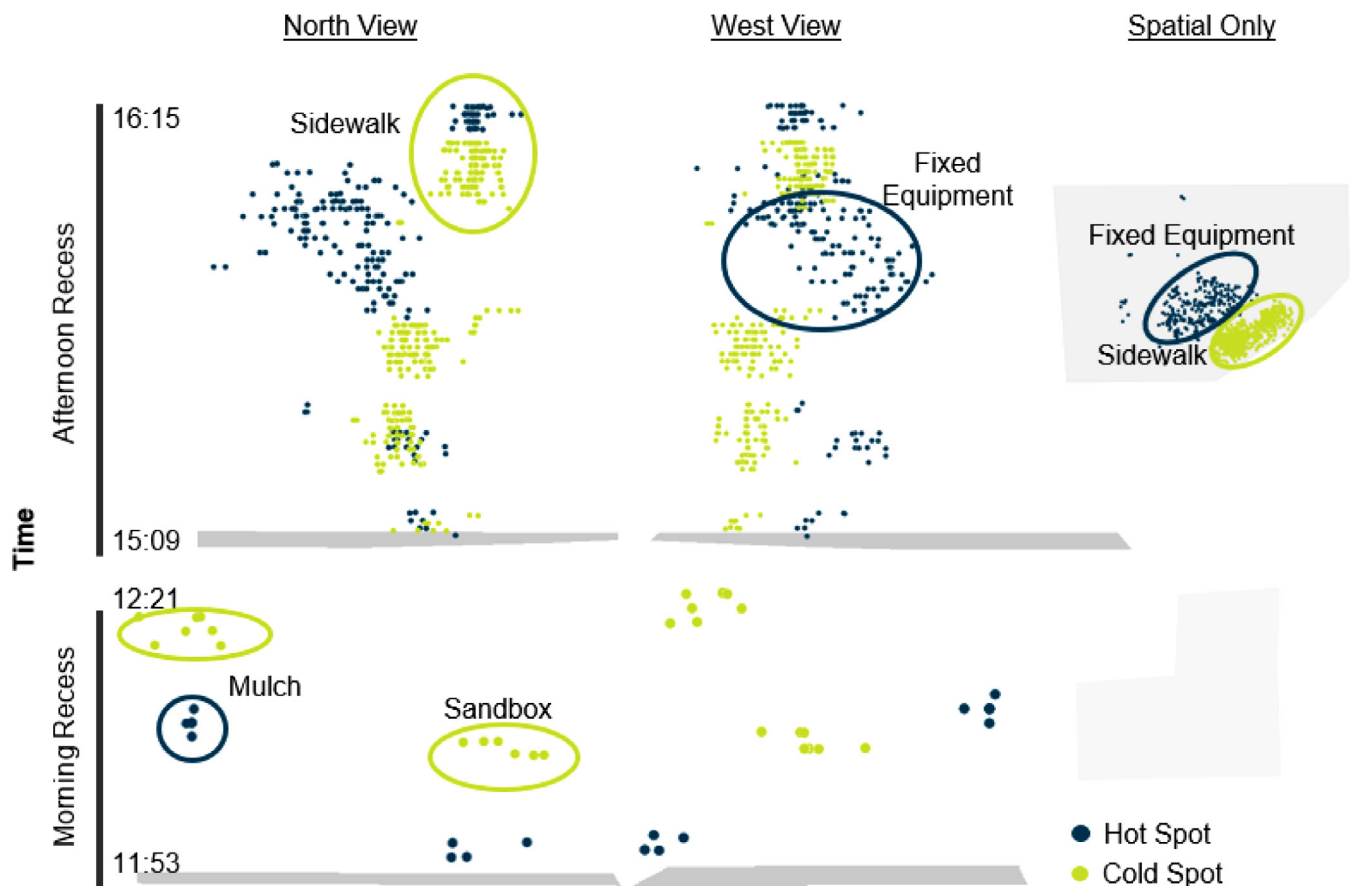


Fig. 4. Significant ($p < 0.05$) spatiotemporal hot and cold spots for classroom 3 on playgrounds 3 (morning) and 4 (afternoon). For the 3-dimensional maps (two left-most panels), the “North View” shows a view from the southern portion of the playground looking North while the “West View” shows a view from the eastern portion of the playground looking West with time shown on the y-axis (further up the y-axis, more time has elapsed). The spatial only model (right-most panel) is shown as an aerial view with North at the top. Figure only shows datapoints belonging to a significant hot or cold spot.

termixed with a cold spot in that same location during the lunch recess period. However, during the afternoon recess, children had a prolonged hot spot, with no cold spots, and vector magnitude was generally high and increasing throughout the outdoor period (Table 2). Although this was not directly investigated in the present study, this may be related to the fact that it had been four hours since the children’s last outdoor break. This is in line with play deprivation theory, which posits that longer confinements result in a rebound effect, whereby children are more active when allowed outdoors (Pellegrini et al., 1995; Smith and Hagan, 1980). This relationship between the timing of different outdoor periods is also evident in the other class (class 2) in the same childcare center. Class 2’s first recess session (“morning recess”) did not occur until after 11:00 am, during which they had no large cold spots, while the class with the morning recess period that occurred immediately after arrival at school had both hot and cold spots during their “lunch recess.” There was only a two-hour deprivation before class 2’s last outdoor break of the day (“afternoon recess 2”), where they demonstrated a more oscillating hot/cold spot pattern and vector magnitude levels (Table 2) compared to class 1’s afternoon recess which was preceded by 4 h of indoor time. While this was not directly studied in the present analysis, these incidences provide support for further investigation of the impact of prior deprivation on children’s behavior during outdoor time.

In contrast to our supposition that longer periods of prior deprivation result in hot spots at the onset of outdoor time, class 3 had a three-hour break between morning and afternoon recess, but still had large cold spots at the onset of their time on play-

ground 4 (Fig. 3). This may be because they had to walk to that playground (approximately 5 min), expending some of their ‘pent up’ energy before arriving at the playground. As this could be related to other factors (e.g., teacher involvement, provided additional equipment, indoor physical activity levels, weather), future researchers should directly explore how the length of deprivation impacts activity levels on the playground. Similarly, some research has suggested that differences in physical activity between outdoor periods (e.g., morning vs. afternoon) may be impacted by seasonal variation (Soini et al., 2014). We posit that the methodology described in this study could be used to explore how where and when children are active is impacted by the outdoor schedule or time of year but regardless of the reason for these temporal trends, our study supports the importance of considering time in addition to physical location.

Researchers have suggested that the length of outdoor time impacts children’s activity level, with longer outdoor periods resulting in lower levels of physical activity (Cardon et al., 2008). In the present study, outdoor time ranged from 18 to 91 min for an individual session, but we did not find that longer outdoor periods elicited lower vector magnitude levels. Similarly, McKenzie et al. (1997) and Holmes (2012) have demonstrated that children tended to be less physically active as the recess period elapsed. However, we found that this trend of declining physical activity throughout the outdoor period was not always present (Table 2), but rather, activity level sometimes increased or did not change. This likely depends on a number of factors, including the length of prior deprivation, provided equipment, or teacher in-

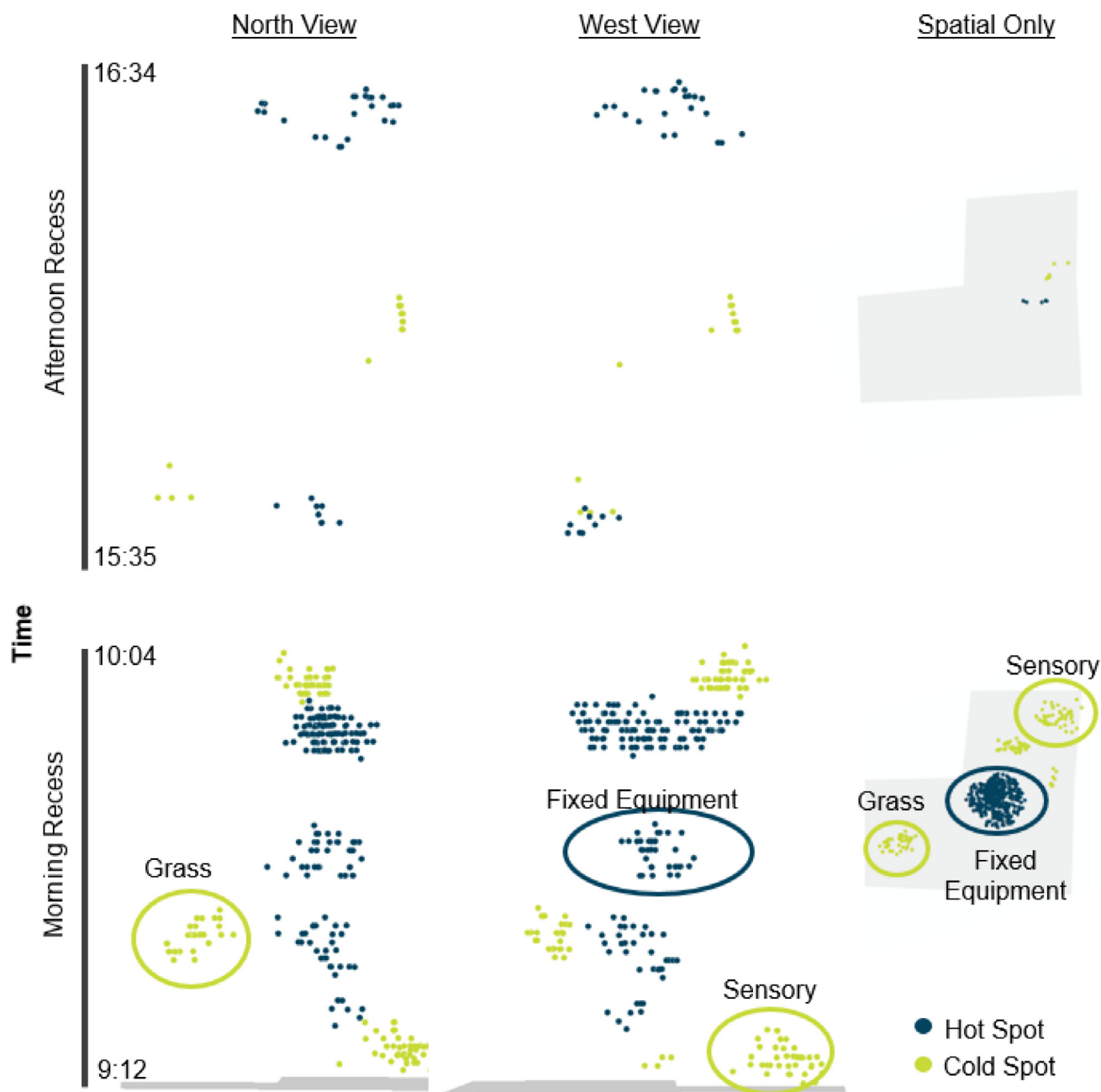


Fig. 5. Significant ($p < 0.05$) spatiotemporal hot and cold spots for classroom 4 on playground 3. For the 3-dimensional maps (two left-most panels), the “North View” shows a view from the southern portion of the playground looking North while the “West View” shows a view from the eastern portion of the playground looking West with time shown on the y-axis (further up the y-axis, more time has elapsed). The spatial only model (right-most panel) is shown as an aerial view with North at the top. Figure only shows datapoints belonging to a significant hot or cold spot.

volvement. Regardless of the physical activity trend, we extend this work by demonstrating that where children were being active also changes over the course of the outdoor period. One important example is fixed equipment, which has been reported to elicit physical activity. However, fixed equipment can also result in cold spots (e.g., Figs. 2 and 6) at certain times during the outdoor period. Future researchers may investigate the timing of these cold spots, identify what children are doing on the fixed equipment during these times, and if warranted, develop intervention strategies to mitigate unwanted behavior. In contrast, there was one playground (Fig. 3) in which the spatial relationships were fairly consistent over time. As can be seen in Fig. 3, the sandbox had a temporally consistent cold spot, while the grass had a consistent hot spot throughout the outdoor period. Thus, in this playground, the

grassy area was able to consistently promote physical activity, a potentially promising point of intervention.

One concern when using hot spot analysis is whether this statistical test is capable of discerning ‘high’ activity counts from ‘low’ activity counts in a meaningful way. When using spatiotemporal weight matrices in place of spatial only models, the difference in vector magnitude between hot and cold spots became more pronounced (Table 3). This may be beneficial because what constitutes ‘hot’ and ‘cold’ will depend on the distribution of the data and if all children are fairly active, then the cold spots may not be sedentary, but rather just less intensely active (i.e., light-to-moderate activity), or if overall activity levels are lower, hot spots may not correspond to moderate-to-vigorous intensity activity. While there is no universally accepted method to classify vector magnitude

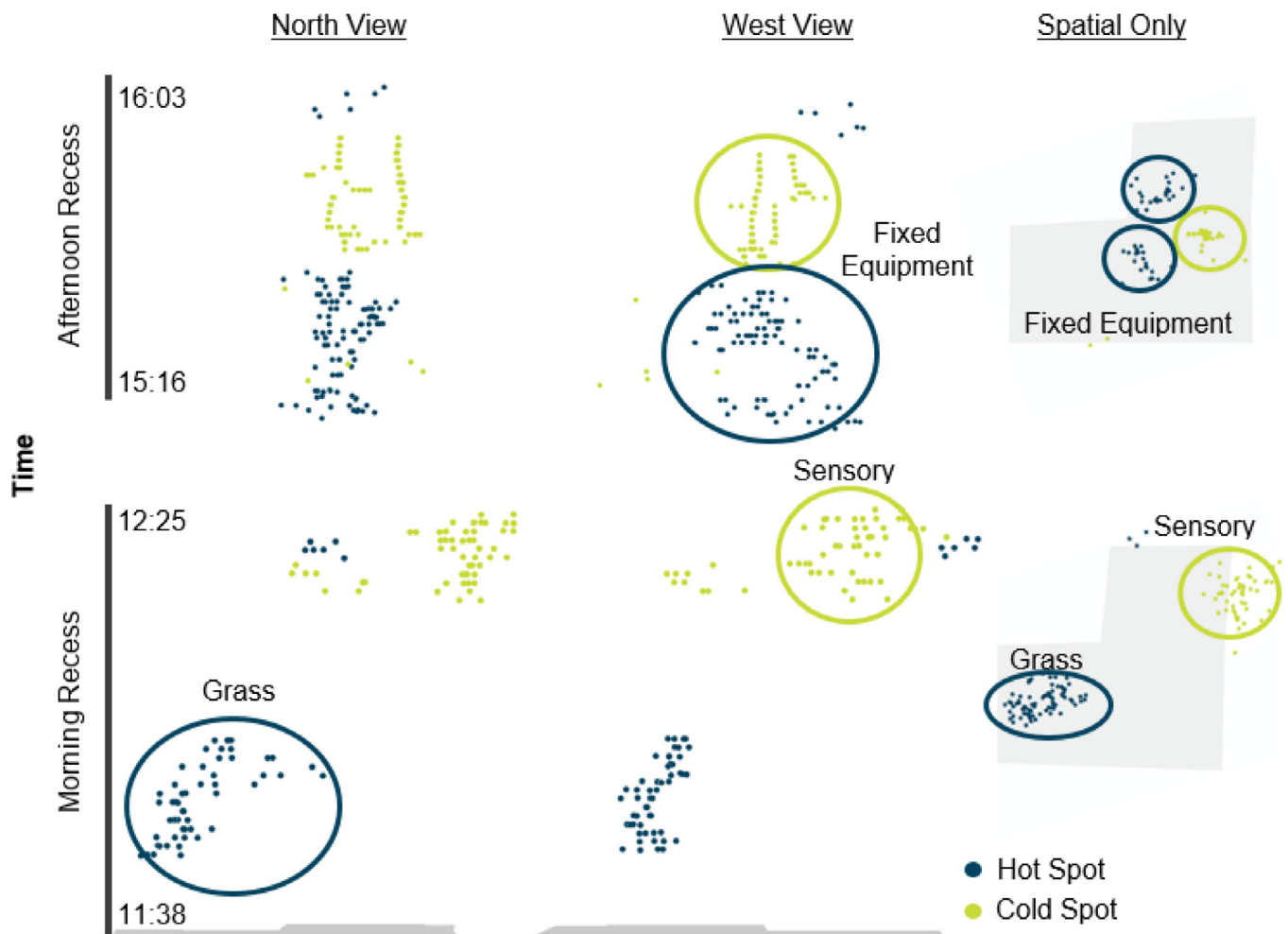


Fig. 6. Significant ($p < 0.05$) spatiotemporal hot and cold spots for classroom 5 on playground 3. For the 3-dimensional maps (two left-most panels), the “North View” shows a view from the southern portion of the playground looking North while the “West View” shows a view from the eastern portion of the playground looking West with time shown on the y-axis (further up the y-axis, more time has elapsed). The spatial only model (right-most panel) is shown as an aerial view with North at the top. Figure only shows datapoints belonging to a significant hot or cold spot.

in 15-s epochs into physical activity levels in preschoolers, a few cut-points exist and can provide a point of reference for interpreting our findings. Butte et al. (2014) classifies sedentary behavior in 3- to 5-year-old children as less than 205 counts/15-s, while 977 and 1528 are cut-points for light-to-moderate activity and moderate-to-vigorous intensity activity, respectively. Additionally, Jimmy et al. (2013) and Costa et al. (2014) define the moderate intensity activity cut-point as 738 and 1086, respectively. Using these cut-point values as a reference point, it is clear that the spatiotemporal approach better separated hot and cold spots, making them better align with sedentary-to-light versus moderate-to-vigorous activity levels. Regardless, caution should be used when interpreting any hot spot analysis and hot and cold spots should be thought of as higher versus lower intensity areas, not necessarily moderate-to-vigorous versus sedentary areas.

The present study was not without limitations. Namely, we focused on two high-quality childcare centers located near a major university, so socioeconomic status and available resources may have been different from other childcare centers. Both childcare centers that participated in this project provided above-average levels of outdoor time (Tandon et al., 2013; Vanderloo et al., 2013). Data collection occurred in the summer, which may have impacted where and how actively children played (e.g., when it is warm, children may have cold spots in the shade toward the end of re-

cess) but future research could use this GPS plus accelerometry approach to understand how weather impacts the timing of hot and cold spots on the playground.

An important consideration is whether hot spot analysis is the most appropriate method for identifying more or less active playground locations. One issue is that each child contributes multiple data points so hot or cold spots may be driven by the presence of a particularly active or inactive child. Use of novel methods for adjusting for trajectories (Nikitopoulos et al., 2018) or recruitment of larger samples may help to address this issue. As this impacts the conclusions that can be drawn from this type of analysis, it is important to note that we cannot conclude that these particular times/locations elicit hot/cold spots in all children. Rather, this study simply points out that clusters locations may vary over time and this should be incorporated in to future studies. There may also be important group differences not revealed when analyzing data from all children at one time, so differences between individual and groups of children should be further studied. In particular, it is of interest to explore whether the same cluster patterns are found for children not meeting physical activity guidelines, girls, or overweight/obese children. This method may also be more appropriate for use in older children and/or adolescents, as the purpose of outdoor time in the preschool age group is not just intense physical activity, but also the development of fine and gross

motor skills, social skills, and for cognitive restoration and development. Finally, more research is needed to better understand the optimal parameters for this type of analysis in this setting (e.g., varying time bands).

4.1. Conclusions

Numerous researchers, teachers, and policy makers are interested in utilizing outdoor time to promote the health and well-being of young children. Prior research has demonstrated that the childcare one attends accounts for up to half of the variability in children's daily physical activity (Gubbels et al., 2011; Pate et al., 2008; Tandon et al., 2013; Vanderloo et al., 2014), irrevocably demonstrating that there are some important contextual or environmental factors within the childcare that impacts children's behavior. Intervention strategies have been developed in an attempt to capitalize on these factors and have included playground redesign (Nicaise et al., 2012), provision of additional outdoor time (Alhassan et al., 2007) or equipment (Cardon et al., 2009), or providing structured, in place of unstructured, outdoor periods (Frank et al., 2018). These approaches have had mixed results, potentially due to our incomplete understanding of how the environment influences children's physical activity behaviors. Use of GPS plus accelerometry in future studies will allow researchers to identify not just where and how actively children play on playgrounds of varying design, or with different outdoor schedules or intervention strategies, but also when these behaviors are occurring (e.g., the start of outdoor time). In the present study, we identified interperiod and intraperiod changes in the location of hot and cold spots, supporting the importance of considering the impact of both spatial and temporal influences on children's playground physical activity levels.

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Declarations of Competing Interest

None

References

- Alhassan, S., Sirard, J.R., Robinson, T.N., 2007. The effects of increasing outdoor play time on physical activity in Latino preschool children. *Int. J. Pediatric Obesity* 2 (3), 153–158.
- Andersen, H.B., Christiansen, L.B., Pawlowski, C.S., Schipperijn, J., 2019. What we build makes a difference—Mapping activating schoolyard features after renewal using GIS, GPS and accelerometers. *Landsc. Urban Plan.* 191, 103617.
- Andersen, H.B., Klinker, C.D., Toftager, M., Pawlowski, C.S., Schipperijn, J., 2015. Objectively measured differences in physical activity in five types of schoolyard area. *Landsc. Urban Plan.* 134, 83–92.
- ArcMap. Optimized Hot Spot Analysis. Retrieved from <http://desktop.arcgis.com/en/arcmap/10.5/tools/spatial-statistics-toolbox/optimized-hot-spot-analysis.htm>
- Berg, S., 2015. Children's activity levels in different playground environments: an observational study in four Canadian preschools. *Early Childhood Educ. J.* 43 (4), 281–287.
- Black, I.E., Menzel, N.N., Bungum, T.J., 2015. The relationship among playground areas and physical activity levels in children. *J. Pediatr. Health Care* 29 (2), 156–168.
- Boldemann, C., Blennow, M., Dal, H., Mårtensson, F., Raustorp, A., Yuen, K., Wester, U., 2006. Impact of preschool environment upon children's physical activity and sun exposure. *Prev. Med.* 42 (4), 301–308.
- Bower, J.K., Hales, D.P., Tate, D.F., Rubin, D.A., Benjamin, S.E., Ward, D.S., 2008. The childcare environment and children's physical activity. *Am. J. Prev. Med.* 34 (1), 23–29.
- Brown, W.H., Pfeiffer, K.A., McIver, K.L., Dowda, M., Addy, C.L., Pate, R.R., 2009. Social and environmental factors associated with preschoolers' nonsedentary physical activity. *Child Dev.* 80 (1), 45–58.
- Butte, N.F., Wong, W.W., Lee, J.S., Adolph, A.L., Puyau, M.R., Zakeri, I.F., 2014. Prediction of energy expenditure and physical activity in preschoolers. *Med. Sci. Sports Exerc.* 46 (6), 1216.
- Caldas de Castro, M., Singer, B.H., 2006. Controlling the false discovery rate: a new application to account for multiple and dependent tests in local statistics of spatial association. *Geogr. Anal.* 38 (2), 180–208.
- Cardon, G., Labarque, V., Smits, D., De Bourdeaudhuij, I., 2009. Promoting physical activity at the pre-school playground: the effects of providing markings and play equipment. *Prev. Med.* 48 (4), 335–340.
- Cardon, G., Van Cauwenberghe, E., Labarque, V., Haerens, L., De Bourdeaudhuij, I., 2008. The contribution of preschool playground factors in explaining children's physical activity during recess. *Int. J. Behav. Nutr. Phys. Activity* 5 (1), 11.
- Clark, J.E., Clements, R.L., Guddemi, M., Morgan, D.W., Pica, R., Pivarnik, J.M., ... Virgilio, S.J., 2002. Active Start: A Statement of Physical Activity Guidelines for Children Birth to Five Years. AAHPERD Publications.
- Clevenger, K.A., Sinha, G., Howe, C.A., 2018. Comparison of methods for analyzing Global Positioning System and accelerometer data during school recess. *Meas. Phys. Educ. Exerc. Sci.* 23 (1), 58–68.
- Commonwealth of Australia, 2010. National Physical Activity Guidelines For Australians. Physical activity Recommendations For 0–5 Year Olds. Commonwealth of Australia, Canberra, Australia.
- Corcoran, L.S., Katrina. (2019). *Early Childhood Program Participation, Results from the National Household Education Surveys Program of 2016*. Retrieved from <https://nces.ed.gov/pubsearch/pubsinfo.asp?pubid=2017101REV>.
- Cosco, N.G., Moore, R.C., Islam, M.Z., 2010. Behavior mapping: a method for linking preschool physical activity and outdoor design. *Med. Sci. Sports Exerc.* 42 (3), 513–519.
- Costa, S., Barber, S.E., Cameron, N., Clemen, S.A., 2014. Calibration and validation of the ActiGraph GT3X+ in 2–3 year olds. *J. Sci. Med. Sport* 17 (6), 617–622.
- Dowda, M., Brown, W.H., McIver, K.L., Pfeiffer, K.A., O'Neill, J.R., Addy, C.L., Pate, R.R., 2009. Policies and characteristics of the preschool environment and physical activity of young children. *Pediatrics* 123 (2), e261–e266.
- Duncan, S., Stewart, T.I., Oliver, M., Mavoa, S., MacRae, D., Badland, H.M., Duncan, M.J., 2013. Portable global positioning system receivers: static validity and environmental conditions. *Am. J. Prev. Med.* 44 (2), e19–e29.
- Fjørtoft, I., Kristoffersen, B., Sageie, J., 2009. Children in schoolyards: Tracking movement patterns and physical activity in schoolyards using global positioning system and heart rate monitoring. *Landsc. Urban Plan.* 93 (3–4), 210–217.
- Fjørtoft, I., Löfman, O., Thorén, K.H., 2010. Schoolyard physical activity in 14-year-old adolescents assessed by mobile GPS and heart rate monitoring analysed by GIS. *Scand. J. Public Health* 38 (5 suppl), 28–37.
- Frank, M.L., Flynn, A., Farnell, G.S., Barkley, J.E., 2018. The differences in physical activity levels in preschool children during free play recess and structured play recess. *J. Exerc. Sci. Fitness* 16 (1), 37–42.
- Getis, A., Ord, J.K., 1992. The analysis of spatial association by use of distance statistics. *Geogr. Anal.* 24 (3), 189–206.
- Gubbels, J.S., Kremers, S.P., Van Kann, D.H., Stafleu, A., Candel, M.J., Dagnelie, P.C., ... De Vries, N.K., 2011. Interaction between physical environment, social environment, and child characteristics in determining physical activity at child care. *Health Psychol.* 30 (1), 84.
- Gunter, K.B., Rice, K.R., Ward, D.S., Trost, S.G., 2012. Factors associated with physical activity in children attending family child care homes. *Prev. Med.* 54 (2), 131–133.
- Holmes, R.M., 2012. The Outdoor Recess Activities of Children at an Urban School: Longitudinal and Intraperiod Patterns. *Am. J. Play* 4 (3), 327–351.
- Howe, C.A., Clevenger, K.A., Plow, B., Porter, S., Sinha, G., 2018. Using video direct observation to assess children's physical activity during recess. *Pediatric Exerc. Sci.* 30 (4), 516–523.
- Hustiy, K.M., Normand, M.P., Larson, T.A., Morley, A.J., 2012. The effect of outdoor activity context on physical activity in preschool children. *J. Appl. Behav. Anal.* 45 (2), 401–405.
- Jimmy, G., Seiler, R., Maeder, U., 2013. Development and validation of energy expenditure prediction models based on GT3X accelerometer data in 5-to 9-year-old children. *J. Phys. Activity Health* 10 (7), 1057–1067.
- Larson, T.A., Normand, M.P., Morley, A.J., Hustiy, K.M., 2014. The role of the physical environment in promoting physical activity in children across different group compositions. *Behav. Modif.* 38 (6), 837–851.
- McKenzie, T.L., Sallis, J.F., Elder, J.P., Berry, C.C., Hoy, P.L., Nader, P.R., ... Broyles, S.L., 1997. Physical activity levels and prompts in young children at recess: a two-year study of a bi-ethnic sample. *Res. Q. Exerc. Sport* 68 (3), 195–202.
- Miguel, J.H., Cadenas-Sanchez, C., Ekelund, U., Nyström, C.D., Mora-Gonzalez, J., Löf, M., ... Ortega, F.B., 2017. Accelerometer data collection and processing criteria to assess physical activity and other outcomes: a systematic review and practical considerations. *Sports Med.* 47 (9), 1821–1845.
- Moran, P.A., 1950. A test for the serial independence of residuals. *Biometrika* 37 (1/2), 178–181.
- Nicaise, V., Kahan, D., Reuben, K., Sallis, J.F., 2012. Evaluation of a redesigned outdoor space on preschool children's physical activity during recess. *Pediatric Exerc. Sci.* 24 (4), 507–518.
- Nicaise, V., Kahan, D., Sallis, J.F., 2011. Correlates of moderate-to-vigorous physical activity among preschoolers during unstructured outdoor play periods. *Prev. Med.* 53 (4), 309–315.

- Nikitopoulos, P., Paraskevopoulos, A.I., Doukeridis, C., Pelekis, N., Theodoridis, Y., 2018, December. Hot spot analysis over big trajectory data. In: 2018 IEEE International Conference on Big Data, pp. 761–770. doi:10.1109/BigData.2018.8622376.
- Oliver, M., Schofield, G.M., Kolt, G.S., 2007. Physical activity in preschoolers: understanding prevalence and measurement issues. *Sports Med.* 37 (12), 1045–1070.
- Pate, R.R., McIver, K., Dowda, M., Brown, W., Addy, C., 2008. Directly observed physical activity levels in preschool children. *J. Sch. Health* 78 (8), 438–444.
- Pate, R.R., Pfeiffer, K.A., Trost, S.G., Ziegler, P., Dowda, M., 2004. Physical activity among children attending preschools. *Pediatrics* 114 (5), 1258–1263.
- Pawlowski, C.S., Andersen, H.B., Troelsen, J., Schipperijn, J., 2016. Children's physical activity behavior during school recess: A pilot study using GPS, accelerometer, participant observation, and go-along interview. *PLoS One* 11 (2), e0148786.
- Pellegrini, A.D., Huberty, P.D., Jones, L., 1995. The effects of recess timing on children's playground and classroom behaviors. *Am. Educ. Res. J.* 32 (4), 845–864.
- QStarz. BT-Q1300ST Specifications. Retrieved from http://www.qstarz.com/Products/GPS_Products/BT-Q1300ST-S.htm
- Saint-Maurice, P., Welk, G.J., Silva, P., Siahpash, M., Huberty, J., 2011. Assessing children's physical activity behaviors at recess: A multi-method approach. *Pediatr. Exerc. Sci.* 23 (4), 585–599.
- Smith, P.K., Hagan, T., 1980. Effects of deprivation on exercise play in nursery school children. *Anim. Behav.* 28 (3), 922–928.
- Smith, W.R., Moore, R., Cosco, N., Wesoloski, J., Danninger, T., Ward, D.S., Ries, N., 2014. Increasing physical activity in childcare outdoor learning environments the effect of setting adjacency relative to other built environment and social factors. *Environ. Behav.* 48 (4), 550–578.
- Soini, A., Villberg, J., Sääkslahti, A., Gubbels, J., Mehtälä, A., Kettunen, T., Poskiparta, M., 2014. Directly observed physical activity among 3-year-olds in Finnish childcare. *Int. J. Early Childhood* 46 (2), 253–269.
- Sugiyama, T., Okely, A.D., Masters, J.M., Moore, G.T., 2012. Attributes of child care centers and outdoor play areas associated with preschoolers' physical activity and sedentary behavior. *Environ. Behav.* 44 (3), 334–349.
- Tandon, P.S., Saelens, B.E., Zhou, C., Kerr, J., Christakis, D.A., 2013. Indoor versus outdoor time in preschoolers at child care. *Am. J. Prev. Med.* 44 (1), 85–88.
- Tremblay, M.S., Chaput, J.-P., Adamo, K.B., Aubert, S., Barnes, J.D., Choquette, L., ... Gray, C.E., 2017. Canadian 24-Hour movement guidelines for the early years (0–4 years): An integration of physical activity, sedentary behaviour, and sleep. *BMC Public Health* 17 (5), 874.
- Trost, S.G., Fees, B., Dziewaltowski, D., 2008. Feasibility and efficacy of a "Move and Learn" physical activity curriculum in preschool children. *J. Phys. Activity Health* 5 (1), 88.
- United Kingdom Chief of Medical Officers. (2011). Start active, stay active: A report on physical activity for health from the four home countries' chief medical officers. Available at https://www.sportengland.org/media/2928/dh_128210.pdf.
- United States Department of Health and Human Services (USDHHS), 2018. Physical Activity Guidelines For Americans, 2nd Edition Available from https://health.gov/paguidelines/second-edition/pdf/Physical_Activity_Guidelines_2nd_edition.pdf.
- Van Cauwenberghe, E., De Bourdeaudhuij, I., Maes, L., Cardon, G., 2012. Efficacy and feasibility of lowering playground density to promote physical activity and to discourage sedentary time during recess at preschool: a pilot study. *Prev. Med.* 55 (4), 319–321.
- Vanderloo, L.M., Tucker, P., Johnson, A.M., Holmes, J.D., 2013. Physical activity among preschoolers during indoor and outdoor childcare play periods. *Appl. Physiol. Nutr. Metabol.* 38 (11), 1173–1175.
- Vanderloo, L.M., Tucker, P., Johnson, A.M., van Zandvoort, M.M., Burke, S.M., Irwin, J.D., 2014. The influence of centre-based childcare on preschoolers' physical activity levels: A cross-sectional study. *Int. J. Environ. Res. Public Health* 11 (2), 1794–1802.
- Willenberg, L.J., Ashbolt, R., Holland, D., Gibbs, L., MacDougall, C., Garrard, J., ... Waters, E., 2010. Increasing school playground physical activity: a mixed methods study combining environmental measures and children's perspectives. *J. Sci. Med. Sport* 13 (2), 210–216.